LLM Trust Score — Evaluation Report

Full 200-question benchmark • judge model: Qwen3.8-27B via Groq • correctness-checker: Qwen3.8-27B via Groq

What was tested

200 questions drawn evenly from three published benchmarks (TruthfulQA, HaluEval, TriviaQA). Each question runs through the full pipeline — the base model answers three times, three independent judge models check each answer, a self-consistency check compares the three answers against each other, and a Trust Score is produced. The final chosen answer is then graded for factual correctness against the benchmark's known answer.

Headline numbers

Metric	Result	What it means
Answer accuracy	**60.0%** (120 / 200)	How often the base model's final answer was actually right. A ceiling set by the base model's knowledge, not the scoring layer.
Avg Trust Score — correct answers	82.6%	When the answer was right, the system was confident.
Avg Trust Score — wrong answers	67.2%	When the answer was wrong, confidence dropped noticeably.
Gap	**15.4 points**	The core result: the Trust Score is meaningfully lower on bad answers than good ones, so it carries real signal.

Does the confidence label mean anything?

Confidence band	# answers	Actually correct	Reading
High (Trust ≥ 80%)	108	69%	Above the 60.0% baseline — “High” leans reliable.
Medium (75–79%)	26	65%	Still above baseline, if less sharply.
Low (Trust < 75%)	66	**44%**	A real signal — a “Low” score is wrong more often than not.

The confidence bands separate cleanly (69% → 65% → 44% as Trust Score drops), though the separation is softer here than with the previous judge models — consistent with the narrower 17.2 → 15.4 point gap trend as the judge model gets smaller.

Accuracy by dataset (this run)

Dataset	Accuracy	Why

TriviaQA	80.9% (55 / 68)	Straightforward factual recall — stays strong regardless of judge model.
TruthfulQA	56.1% (37 / 66)	Designed to bait common human misconceptions.
HaluEval	42.4% (28 / 66)	Hardest by design — multi-hop questions chaining several obscure facts. A base-knowledge limit, not a scoring flaw.

Per-dataset accuracy across all 3 judge models tried

Dataset	Claude Haiku (original)	GPT-OSS-20B	Qwen3.8-27B (this run)
TruthfulQA	68.2%	60.6%	56.1%
HaluEval	50.0%	47.0%	42.4%
TriviaQA	79.4%	79.4%	80.9%

Pattern: TriviaQA (simple factual recall) stays rock-steady around 79–81% regardless of which model judges it — that task is easy for any of the three judges. TruthfulQA and HaluEval both trend downward as the judge model gets smaller (Claude Haiku → GPT-OSS-20B → Qwen3.8-27B), since both datasets hinge on catching subtler misconceptions or multi-hop reasoning that a smaller judge is more likely to misjudge on borderline cases. HaluEval is the hardest dataset across all three judges, as expected by design.

Overall comparison across all 3 configurations

	Claude Haiku (original)	GPT-OSS-20B	Qwen3.8-27B (this run)
Overall accuracy	66.0%	62.5%	60.0%
Trust Score gap	23.0 pts	17.2 pts	15.4 pts

Qwen3.8-27B is the least sharp discriminator of the three judge models tried — both accuracy and the trust-score gap are lowest here — but the system remains clearly functional, with a solid positive 15.4-point separation between confident-correct and confident-wrong answers.

Full per-question data: evaluation/results_final.csv • Confidence thresholds: High ≥ 80%, Medium ≥ 75%, Low < 75%.